Data Report
Lithology Description, TOC and Rock-Eval of 36 DC Samples from 6406/2-2, 6406/2-5 and 15/9-19 B T2

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Table 1. Number of analyses performed

AnAnalysis	Cuttings	Cuttings, pre-extracted	Total
Lithology	36		36
Pre-extraction	36		36
TOC/Rock-Eval	36	36	72

Table 2. Lithology Description

Well	Sample type	Lower Depth	APT ID	%	Lithology	Attributes
6406/2-2	DC	4355	18628A	30%	CLYST	rdbrn calc
6406/2-2	DC	4355	18628B	50%	CLYST	dkgry-blk, non calc
6406/2-2	DC	4355	18628C	20%	CLYST	medgry, silty, sl-calc
6406/2-2	DC	4360	18629A	30%	CLYST	rdbrn aa
6406/2-2	DC	4360	18629B	50%	CLYST	dkgry-blk aa
6406/2-2	DC	4360	18629C	20%	CLYST	medgry silty aa
6406/2-2	DC	4365	18630A	20%	CLYST	rdbrn
6406/2-2	DC	4365	18630B	60%	CLYST	medgry-ltgry
6406/2-2	DC	4365	18630C	20%	CLYST	medgry
6406/2-2	DC	4370	18631A	70%	CLYST	medgry silty
6406/2-2	DC	4370	18631B	25%	CLYST	med-dkgry
6406/2-2	DC	4370	18631C	5%	CLYST	blk
6406/2-2	DC	4370	18631D	trace	LST	
6406/2-2	DC	4370	18631E	trace	PYR	
6406/2-2	DC	4375	18632A	95%	CLYST	blk
6406/2-2	DC	4375	18632B	5%	CLYST	med-ltgry
6406/2-2	DC	4375	18632C	trace	PYR	
6406/2-2	DC	4380	18633	100 %	CLYST	blk
6406/2-2	DC	4385	18634A	95%	CLYST	blk
6406/2-2	DC	4385	18634B	5%	CLYST	medgry, dkgry, rd-brn
6406/2-2	DC	4385	18634C	trace	PYR	
6406/2-2	DC	4390	18635A	80%	CLYST	med-dkgry
6406/2-2	DC	4390	18635B	15%	CLYST	blk
6406/2-2	DC	4390	18635C	5%	CLYST	med-ltgry
6406/2-2	DC	4395	18636	100 %	CLYST	med-dkgry
6406/2-2	DC	4400	18637A	80%	CLYST	med-dkgry
6406/2-2	DC	4400	18637B	20%	CLYST	dkgry-blk
6406/2-2	DC	4405	18638	100 %	CLYST	med-dkgry
6406/2-5	DC	4650	18639A	40%	CLYST	medgry, medgry-ltgry
6406/2-5	DC	4650	18639B	40%	CLYST	dkgry
6406/2-5	DC	4650	18639C	20%	SD	ang, clear ??addit
6406/2-5	DC	4653	18640A	80%	CLYST	medgry
6406/2-5	DC	4653	18640B	15%	CLYST	dkgry-blk
6406/2-5	DC	4653	18640C	5%	SD	aa
6406/2-5	DC	4656	18641A	90%	CLYST	medgry
6406/2-5	DC	4656	18641B	10%	CLYST	dkgry
6406/2-5	DC	4656	18641C	trace	SD	ang, clear ??addit
6406/2-5	DC	4659	18642A	80%	CLYST	dkgry, dkgry-blk
6406/2-5	DC	4659	18642B	20%	CLYST	medgry
6406/2-5	DC	4662	18643A	50%	CLYST	medgry-ltgry
6406/2-5	DC	4662	18643B	40%	CLYST	ltgry
6406/2-5	DC	4662	18643C	10%	CLYST	dkgry-blk
6406/2-5	DC	4665	18644A	70%	CLYST	blk
6406/2-5	DC	4665	18644B	20%	CLYST	ltgry
6406/2-5	DC	4665	18644C	10%	SD	ang, clear ??addit
6406/2-5	DC	4668	18645A	90%	CLYST	dkgry-blk
6406/2-5	DC	4668	18645B	10%	CLYST	ltgry sl-calc
6406/2-5	DC	4668	18645C	trace	LST	

Well	Sample type	Lower Depth	APT ID	%	Lithology	Attributes
6406/2-5	DC	4671	18646A	80%	CLYST	dkgry-blk
6406/2-5	DC	4671	18646B	20%	CLYST	ltgry-gngry
6406/2-5	DC	4674	18647A	90%	CLYST	dkgry-blk
6406/2-5	DC	4674	18647B	10%	LST	
6406/2-5	DC	4677	18648A	90%	CLYST	dkgry-blk
6406/2-5	DC	4677	18648B	10%	LST	wh
6406/2-5	DC	4680	18649A	95%	CLYST	dkgry-blk
6406/2-5	DC	4680	18649B	5%	CLYST	ltgry
6406/2-5	DC	4683	18650A	95%	CLYST	dkgry-blk
6406/2-5	DC	4683	18650B	3%	CLYST	ltgry
6406/2-5	DC	4683	18650C	2%	LST	wh
6406/2-5	DC	4686	18651A	50%	CLYST	dkgry-blk
6406/2-5	DC	4686	18651B	35%	CLYST	dkgry
6406/2-5	DC	4686	18651C	15%	CLYST	medgry-brngry silty
6406/2-5	DC	4689	18652A	70%	CLYST	dkgry-blk
6406/2-5	DC	4689	18652B	15%	CLYST	medgry,M-ltgry
6406/2-5	DC	4689	18652C	15%	SD	ang, clear
6406/2-5	DC	4689	18652D	trace	LST	
6406/2-5	DC	4692	18653A	70%	CLYST	medgry-dkgry
6406/2-5	DC	4692	18653B	20%	CLYST	dkgry, dkgry-blk
6406/2-5	DC	4692	18653C	10%	CLYST	ltgry
6406/2-5	DC	4695	18654A	80%	CLYST	medgry
6406/2-5	DC	4695	18654B	20%	CLYST	ltgry, dkgry, blk
6406/2-5	DC	4698	18655A	80%	CLYST	medgry, silty
6406/2-5	DC	4698	18655B	15%	CLYST	dkgry, blk, ltgry
6406/2-5	DC	4698	18655C	5%	SD	ang, clear
6406/2-5	DC	4698	18655D	trace	PYR	
6406/2-5	DC	4701	18656A	90%	CLYST	dkgry, dkgry-blk, blk
6406/2-5	DC	4701	18656B	10%	SD	ang, clear
6406/2-5	DC	4704	18657A	80%	CLYST	dkgry, dkgry-blk
6406/2-5	DC	4704	18657B	10%	SD	ang, clear
6406/2-5	DC	4704	18657C	10%	CLYST	medgry, ltgry
6406/2-5	DC	4707	18658A	80%	CLYST	dkgry, dkgry-blk
6406/2-5	DC	4707	18658B	20%	SD	ang, clear
6406/2-5	DC	4710	18659A	80%	CLYST	dkgry-blk, coaly?
6406/2-5	DC	4710	18659B	trace	SD	ang, clear
15/9-19 B T2	DC	3760	18660A	40%	CLYST	rd-brn, calc
15/9-19 B T2	DC	3760	18660B	40%	CLYST	ltgry, calc
15/9-19 B T2	DC	3760	18660C	20%	CLYST	medgry, calc
15/9-19 B T2	DC	3760	18660D	trace	CLYST	dkgry
15/9-19 B T2	DC	3770	18661A	80%	CLYST	ltgry, calc
15/9-19 B T2	DC	3770	18661B	20%	CLYST	rd-brn, calc
15/9-19 B T2	DC	3780	18662A	40%	CLYST	medgry-dkgry, non-calc
15/9-19 B T2	DC	3780	18662B	30%	CLYST	ltgry, calc
15/9-19 B T2	DC	3780	18662C	30%	CLYST	rd-brn, calc
15/9-19 B T2	DC	3790	18663A	90%	CLYST	dkgry
15/9-19 B T2	DC	3790	18663B	10%	CLYST	ltgry, rd-brn, medgry

Table 3a. TOC and Rock-Eval data

Well	Sample type	Lower Depth	APT ID	S1 (mg/g)	S2 (mg/g)	Tmax (°C)	PP (mg/g)	PI (wt ratio)	HI (mg HC/g TOC)	TOC (%)
6406/2-2	DC	4355	18628B	33.30	7.20	444	40.50	0.82	115	6.26
6406/2-2	DC	4360	18629B	34.77	8.13	449	42.89	0.81	151	5.40
6406/2-2	DC	4365	18630C	33.66	5.72	417	39.38	0.85	146	3.93
6406/2-2	DC	4370	18631C	33.99	10.61	447	44.60	0.76	143	7.43
6406/2-2	DC	4375	18632A	36.61	9.33	446	45.94	0.80	113	8.24
6406/2-2	DC	4380	18633	33.83	8.24	443	42.07	0.80	107	7.72
6406/2-2	DC	4385	18634A	28.97	8.36	452	37.33	0.78	111	7.55
6406/2-2	DC	4390	18635B	31.38	8.03	454	39.40	0.80	105	7.61
6406/2-2	DC	4395	18636	29.90	6.64	439	36.53	0.82	144	4.60
6406/2-2	DC	4400	18637B	34.02	7.68	437	41.70	0.82	151	5.07
6406/2-2	DC	4405	18638	34.89	6.76	424	41.66	0.84	147	4.61
6406/2-5	DC	4650	18639B	35.38	10.97	432	46.35	0.76	187	5.88
6406/2-5	DC	4653	18640B	41.47	11.62	438	53.09	0.78	209	5.56
6406/2-5	DC	4656	18641B	37.34	9.37	430	46.70	0.80	203	4.62
6406/2-5	DC	4659	18642A	37.00	7.26	433	44.26	0.84	179	4.06
6406/2-5	DC	4662	18643C	33.47	8.03	434	41.51	0.81	202	3.97
6406/2-5	DC	4665	18644A	42.37	16.23	447	58.60	0.72	161	10.07
6406/2-5	DC	4668	18645A	49.08	16.16	448	65.23	0.75	153	10.56
6406/2-5	DC	4671	18646A	42.97	16.12	445	59.08	0.73	147	10.93
6406/2-5	DC	4674	18647A	44.10	14.04	450	58.14	0.76	134	10.51
6406/2-5	DC	4677	18648A	37.78	13.19	454	50.97	0.74	126	10.50
6406/2-5	DC	4680	18649A	33.54	11.86	446	45.41	0.74	128	9.24
6406/2-5	DC	4683	18650A	36.53	10.57	440	47.10	0.78	121	8.71
6406/2-5	DC	4686	18651A	36.67	12.91	446	49.58	0.74	143	9.03
6406/2-5	DC	4689	18652A	36.70	10.52	435	47.22	0.78	130	8.09
6406/2-5	DC	4692	18653B	40.34	11.82	434	52.16	0.77	137	8.65
6406/2-5	DC	4695	18654A	43.69	11.39	432	55.08	0.79	154	7.39
6406/2-5	DC	4698	18655B	40.33	12.63	435	52.96	0.76	182	6.94
6406/2-5	DC	4701	18656A	38.63	10.57	434	49.20	0.79	184	5.76
6406/2-5	DC	4704	18657A	38.65	11.37	435	50.01	0.77	212	5.35
6406/2-5	DC	4707	18658A	44.08	12.04	439	56.12	0.79	210	5.74
6406/2-5	DC	4710	18659A	43.72	10.71	435	54.44	0.80	194	5.52
15/9-19 B T2	DC	3760	18660C	26.46	9.51	333	35.97	0.74	277	3.43
15/9-19 B T2	DC	3770	18661A	27.95	8.94	385	36.89	0.76	252	3.54
15/9-19 B T2	DC	3780	18662A	37.84	49.87	428	87.71	0.43	485	10.28
15/9-19 B T2	DC	3790	18663A	45.00	56.59	429	101.59	0.44	448	12.64

Table 3b. TOC and Rock-Eval data after pre- ekstraction

Well	Sample type	Lower Depth	APT ID	S1 (mg/g)	S2 (mg/g)	Tmax (°C)	PP (mg/g)	PI (wt ratio)	HI (mg HC/g TOC)	TOC (%)
6406/2-2	DC, extr.	4355	18628B	0.05	2.64	452	2.69	0.02	167	1.58
6406/2-2	DC, extr.	4360	18629B	0.04	1.91	452	1.95	0.02	219	0.87
6406/2-2	DC, extr.	4365	18630C	0.03	0.89	431	0.92	0.03	168	0.53
6406/2-2	DC, extr.	4370	18631C	0.08	6.54	453	6.62	0.01	122	5.34
6406/2-2	DC, extr.	4375	18632A	0.13	5.48	450	5.61	0.02	112	4.89
6406/2-2	DC, extr.	4380	18633	0.1	4.84	453	4.95	0.02	106	4.59
6406/2-2	DC, extr.	4385	18634A	0.13	4.95	454	5.08	0.03	95	5.22
6406/2-2	DC, extr.	4390	18635B	0.06	2.31	453	2.37	0.03	111	2.08
6406/2-2	DC, extr.	4395	18636	0.06	2.17	448	2.23	0.03	135	1.6
6406/2-2	DC, extr.	4400	18637B	0.07	2.58	448	2.64	0.02	141	1.83
6406/2-2	DC, extr.	4405	18638	0.07	2.34	433	2.42	0.03	189	1.24
6406/2-5	DC, extr.	4650	18639B	0.06	2.62	433	2.68	0.02	305	0.86
6406/2-5	DC, extr.	4653	18640B	0.07	3.88	436	3.95	0.02	315	1.23
6406/2-5	DC, extr.	4656	18641B	0.07	3.11	430	3.17	0.02	437	0.71
6406/2-5	DC, extr.	4659	18642A	0.11	3.2	435	3.32	0.03	493	0.65
6406/2-5	DC, extr.	4662	18643C	0.09	3.58	435	3.67	0.02	560	0.64
6406/2-5	DC, extr.	4665	18644A	0.08	7.87	455	7.95	0.01	139	5.67
6406/2-5	DC, extr.	4668	18645A	0.08	4.8	444	4.88	0.02	132	3.63
6406/2-5	DC, extr.	4671	18646A	0.15	5.17	446	5.32	0.03	89	5.83
6406/2-5	DC, extr.	4674	18647A	0.12	8.35	452	8.46	0.01	131	6.36
6406/2-5	DC, extr.	4677	18648A	0.14	6.44	456	6.59	0.02	93	6.92
6406/2-5	DC, extr.	4680	18649A	0.15	5.57	456	5.72	0.03	94	5.93
6406/2-5	DC, extr.	4683	18650A	0.17	5.19	449	5.36	0.03	105	4.96
6406/2-5	DC, extr.	4686	18651A	0.15	4.99	444	5.14	0.03	86	5.79
6406/2-5	DC, extr.	4689	18652A	0.14	4.51	446	4.65	0.03	104	4.35
6406/2-5	DC, extr.	4692	18653B	0.08	3.62	448	3.71	0.02	120	3.03
6406/2-5	DC, extr.	4695	18654A	0.14	3.9	436	4.04	0.03	174	2.24
6406/2-5	DC, extr.	4698	18655B	0.37	5.44	436	5.81	0.06	222	2.45
6406/2-5	DC, extr.	4701	18656A	0.14	4.83	438	4.97	0.03	239	2.02
6406/2-5	DC, extr.	4704	18657A	0.22	5.22	435	5.44	0.04	328	1.59
6406/2-5	DC, extr.	4707	18658A	0.08	4.57	436	4.65	0.02	303	1.51
6406/2-5	DC, extr.	4710	18659A	0.11	3.93	432	4.04	0.03	291	1.35
15/9-19 B T2	DC, extr.	3760	18660C	0.03	0.6	426	0.63	0.05	154	0.39
15/9-19 B T2	DC, extr.	3770	18661A	0.17	1.5	427	1.67	0.1	283	0.53
15/9-19 B T2	DC, extr.	3780	18662A	0.08	30.84	427	30.92	0	616	5.01
15/9-19 B T2	DC, extr.	3790	18663A	0.12	49.42	425	49.54	0	599	8.25

Experimental Procedures

All procedures follow NIGOGA, 4th Edition. Below are brief descriptions of procedures/analytical conditions.

Sample preparation
Cuttings samples are washed in water to remove mud. When oil based mud is used, soap (Zalo) is added to the sample and the sample is washed thoroughly in warm water to remove mud and soap.

Extraction
A Soxtec Tecator instrument is used. Thimbles are pre extracted in dichloromethane with 7% (vol/vol) methanol, 10 min boiling and 20 min rinsing. The crushed sample is weighed accurately in the pre extracted thimbles and boiled for 1 hour and rinsed for 2 hours in approximately 80 cc of dichloromethane with 7% (vol/vol) methanol. Copper blades activated in concentrated hydrochloric acid are added to the extraction cups to cause free sulphur to react with the copper. An aliquot of 10% of the extract is transferred to a pre weighed bottle and evaporated to dryness. The amount of extractable organic matter is calculated from the weight of this 10% aliquot.

TOC and Rock-Eval
A Rock-Eval 6 instrument is used. The analysis is performed in two steps, pyrolysis and oxidation, when TOC is measured. Jet-Rock 1 was run as every tenth sample and checked against the acceptable range given in NIGOGA.

Temperature programme
Pyrolysis: 300 °C (3 min.) - 25 °C/min. - 650 °C (0 min.)
Oxidation: 400 °C (3 min.) - 25 °C/min. - 850 °C (5 min.)